



No screen. Be seen.

Ross Davenport
Arancha Ramirez
Yasmin Zaraket

Introduction

Overview of project brief and research plan

HMW ensure cyclists safety by improving the navigation and communication system with other vehicles?

Project Brief

Our brief challenged us to develop a product to ensure the physical wellbeing of cyclists by improving the communication system between other road users and cyclists.



Awareness

Key Problems

Removing distractions and simplifying the experience.



Visibility

Be seen easily with minimal change to the activity.

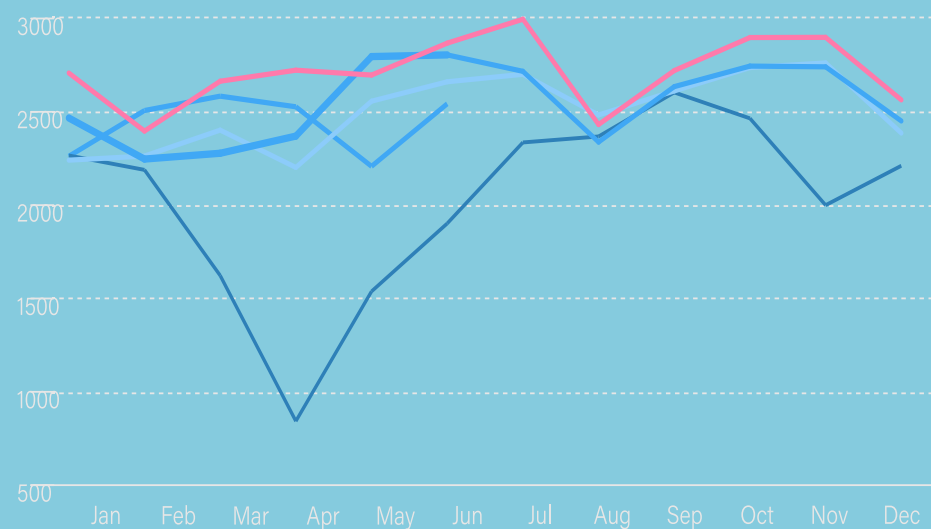


Communication

Improve relations with other road users.

Research Background

- Legislations: Current safety legislations burden cyclists with more responsibilities than their road-sharing counterparts and lessens the desired quality of the cycling experience.
- Casualties: Initial research into cycling safety highlighted that there are still many road deaths and injuries every year involving cyclists, highlighted by the graph showing consistently high casualties below.



Cycling Casualties by Year for London

Target Audience

Based on this findings we decided to target students who lead busy, active lifestyles and commute daily

Interview Insights

- Not all bike lanes are well designed for cyclists.
- Cyclists value their environment.
- Visibility is a major concern when cycling.
- Communication can be hard when cycling.



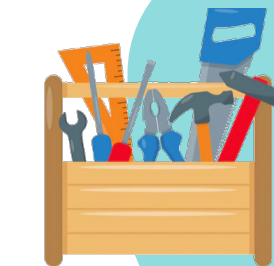
1.Desk research:

Secondary research to determine the importance of product, current legislations aiding cyclists and cycling safety



2.Market research:

Benchmarked current available products on the market targeted to individuals who want to increase their safety whilst cycling. this allowed us to compare and improve our idea and place it in the market.



3.Lo-fi Prototyping:

Lo-fi prototypes will be made aiming to explore the interaction and opinions of a user of the prototype.



4.Co-Design Sessions:

A co-design session will be used to further develop and validate the idea. this will focus on the user input via hands on prototyping and development.

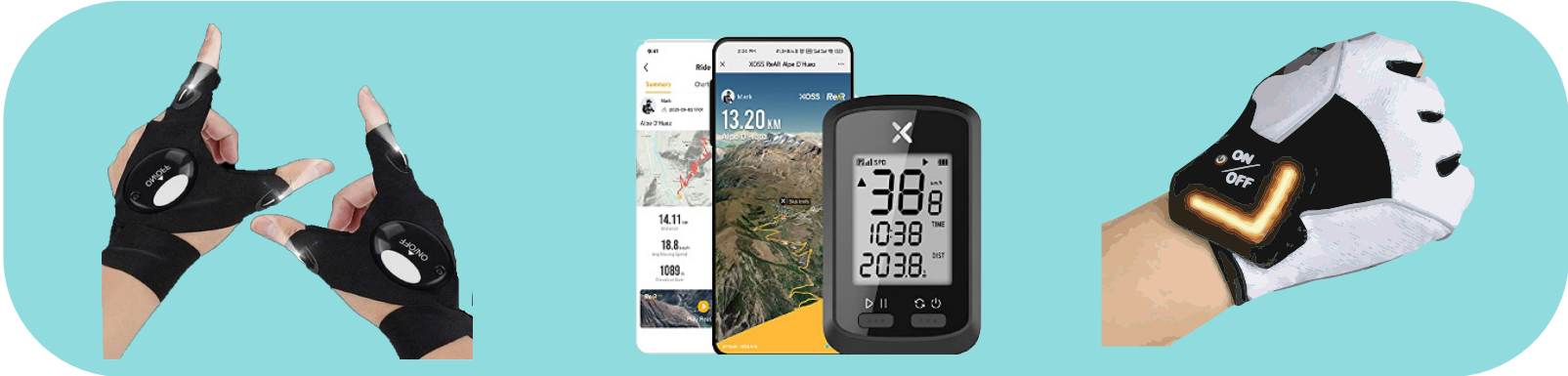


5.Final Concept:

Based on interviews and user feedback, refine idea taking into consideration the user's interaction with final product.

Market Research

Benchmarking current products to identify user’s needs



Name & Price

LED gloves with waterproof lights
Price: £9.95
Website: [Ozerty](#)

XOSS G GPS cycling computer wireless
Price: £28.99
Website: [Amazon](#)

Higolet™ Bicycle Gloves + Turn Signals
Price: £39.99
Website: [Higolet](#)

Description

A button behind the thumb and index joint controls battery-powered LED lights to improve visibility. Activation requires use of other hand.

This device connects to the cyclist’s phone and acts as a speedometer, odometer, and a GPS cycling tracker.

This glove’s LED light activates when the thumb is pressed against the index finger. This brings together the pads and triggers on the light.

Main Insight from Product

Signal Lights on index finger

Navigation system for cyclists

Light activation by pressing thumb and index finger together

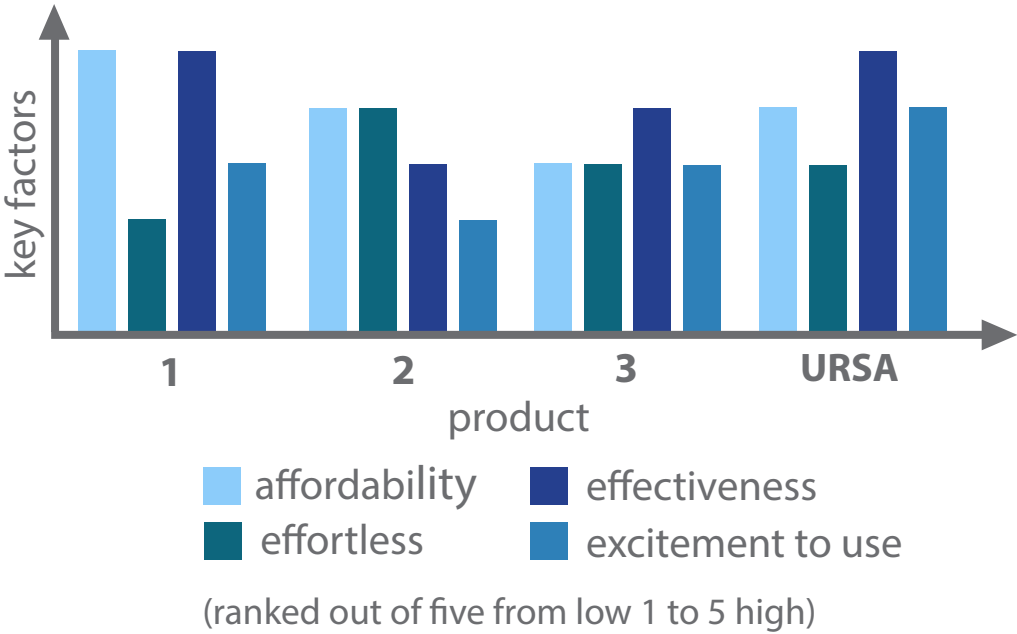
Advantage of URSA

URSA has touch activated lights. No need for buttons.

Navigation via vibration integrated into the glove.

Additional lights on the glove and an inbuilt navigation system.

BENCHMARKING & PRODUCT COMPARISON
Additional research was carried out to identify where our product would fit in the current market



User research

Survey interviews and user feedback

Interviews feedback

Question

What do you believe distracts you the most during your journeys?

Answer

“When I’m looking down at my phone and listening to music.”

Question

How safe do you feel when making a turn? Do you feel visible to other vehicles?

Answer

“It feels more dangerous at night because I can’t be seen as easily.”

User Surveys

We collected additional data with through anonymous surveys that targeted cyclists students between the ages 18-20. The purpose was to identify the user’s safety measures and the biggest fears they encounter in their journeys.

Key Statistics

wears helmet protection



61%

wears additional protection



22%

rely on phone map to travel



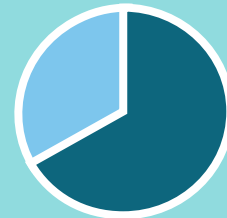
66%

fears making turns



57%

fears entering & exiting roundabouts



66%

Jason Grayson



Age: 25

Occupation: PHD Student at Imperial College London

Mode of Commute: Cycling

Daily Routine:

- Cycle to university every morning (20 mins)
- Protective gear: helmet, rear and front lights, sometimes a reflective jacket
- Route passes through a dangerous roundabout
- Cycle back home after uni (20 mins)
- Sometimes would go on a leisure cycling ride through a park (+10mins)
- Carries bike up three floors to keep in his flat (lift is too narrow for the bike to fit)

“The government should invest more into eco-friendly commute. Maybe even ban all private vehicles from Central London. That way, there’s less traffic and it’s better for the environment.”

Requirements

User needs:

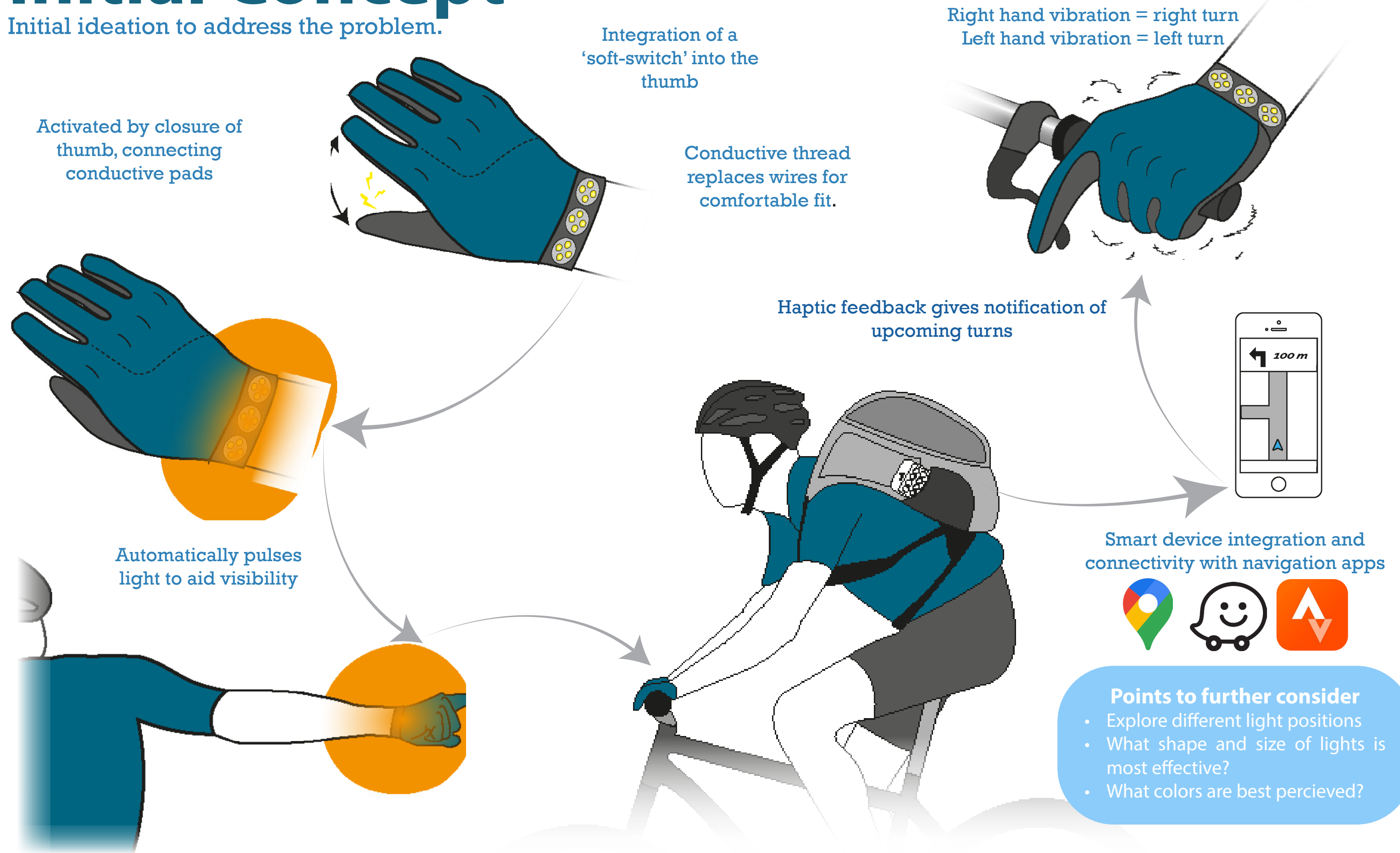
- Simplistic design and straightforward use
- Remain safe whilst cycling
- Being able to arrive at destination without having to take eyes off the road
- Maintain focus whilst cycling, especially whilst travelling through busy roads

User wants:

- Portable and lightweight product
- To relax and enjoy their journey
- Ability to personalize experience

Initial Concept

Initial ideation to address the problem.



Concept Development

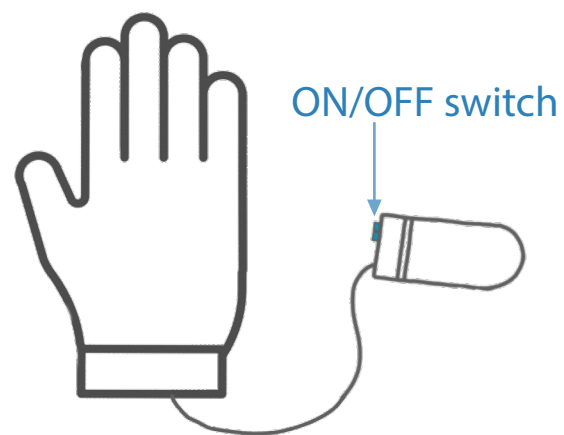
User research and feedback to further define the products features

HOW DOES URSA WORK?

Simply put on the gloves before starting your journey:

1. Turn gloves on, check lights work correctly and connect them to your pre-selected route through the app.
2. Let the vibration guide you to your destiny by indicating turns without needing to take your eyes from the road.
3. When making a turn, extend and close your hand to activate the lights and make a turn safely.
4. When journey concludes, turn off gloves and store away.

EXTRA FEATURES

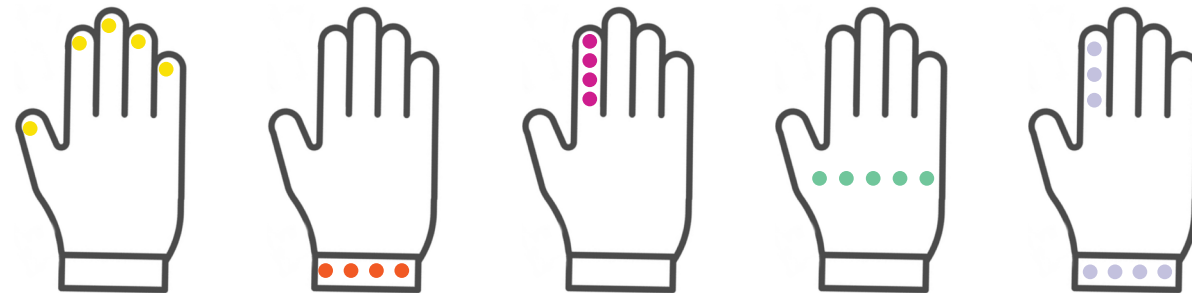


We ended up implementing an ON/OFF switch due to the positive feedback it recieved. Some users pointed out that allowing to disable different features if they became to overwhelming would be very useful.

LIGHT POSITION AND COLOR



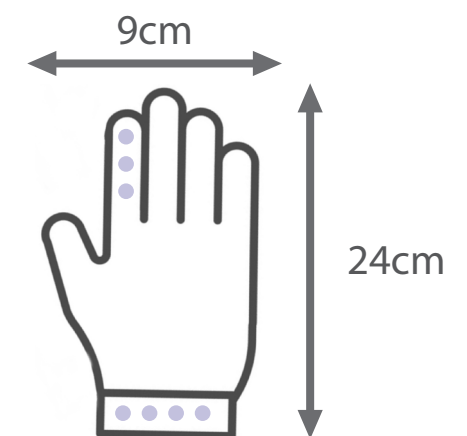
We ideated and brainstormed with different colors in the CAD rendering, but found out that a white bright LED color was best visualized by the users as the rest where too dim or could be easily confused with other vehicles.



Additionally we arranged different light patterns, but found out that a combination between option 2 & 3 received the best feedback as it further reiterates the direction the user is taking, so we combined them into the final pattern (option 5)

INITIAL CAD MODEL AND DIMENSIONS

Some initial rendering and modelling was carried out to further develop the concept



designed for medium sized hand

Lofi Prototyping

Initial prototype, answering preliminary questions, & development

Research Questions:

- What light shape is most comfortable?
- Do the lights position impede movement?
- How do different light positions improve the aesthetics of the glove?
- What is the most comfortable hand movement to activate lights?
- Does the position of the activation system improve the aesthetics?

Points to further consider

- Where should the vibration system be positioned?
- Where would the battery be stored to avoid obstructing movement?
- Will the light in the glove be visible enough?

Key changes

- Light shapes need to be small to not obstruct hand movement
- Activation system between thumb and index finger

User Feedback



Prototyping a long strip of LED lights to increase light visibility:

- The long strip of LEDs difficults movement and flexibility of the hand, making it uncomfortable to grip the handles correctly.
- LED lights positioned evenly from the index finger to the wrist to increase its visibility.



Alternate LED lights position and round shape:

- The smaller, round, less noticeable lights are more aesthetic and provide a sharp finish
- LED lights on the wrist don't obstruct fingers flexibility and still remain visible



Multiple hand movements where prototyped and tested to investigate the most comfortable movement:

- Option 1 was the simplest hand movement as the thumb has a better range of movement. It doesn't even require to fully close all fingers in order to activate the lights.
- Option 1 for the activation system is subtle and barely noticeable, improving URSA's general aesthetic



Works-Like Prototype

Developing a functional and sellable product

Research Questions:

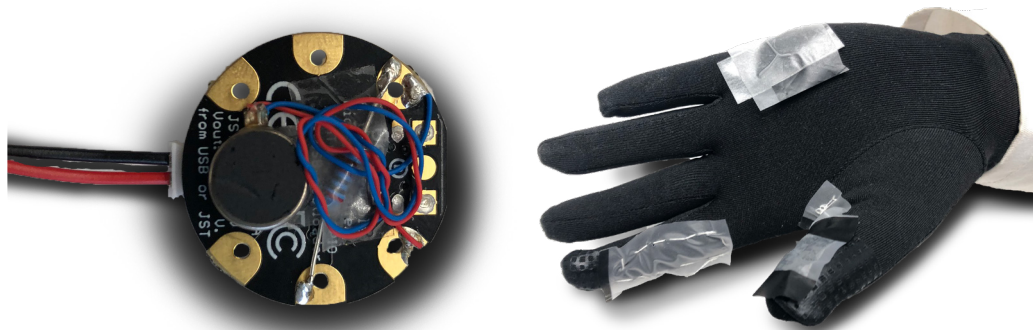
- Where is the vibration best felt?
- Will the vibration position affect the user's movement?
- Will the battery obstruct the vibration?
- Will the battery affect hand's mobility?

Points to further consider

- Does the light activation system accomodate every user's preference?
- Making sure conductive thread doesn't cause a short circuit
- How could the durability of the product be increased?

Key changes

- Add multiple light activation switches to include multiple users
- Straight, simple stitching to avoid short circuits
- Increase vibration
- Reduce space occupied by circuit to reduce fragility of product



User Feedback



Comparing vibration feeling on different places on the wrist:

- The lower vibration position on the wrist or the middle of palm would obstruct movement needed when gripping the handles.
- Situating the vibration system near the lower back of the hand will still be effective and unobstructive as the hand and wrist still have full mobility.
- If the vibration is strong enough it will be felt, even if it's not fully touching the skin

Battery and circuit positioned in different places to assess glove mobility:

- Battery tightly stored on the cuff of the glove doesn't restrict cyclist wrist movement

Using arduino we programmed the LED lights to light up when the user closed its thumb and index finger which activates the circuit.

Additionally, the vibration system was programmed to simulate a signal from the hypothetical navigation app

```
button_test | Arduino 1.8.19
Edit Sketch Tools Help

button_test
// variables will change:
int buttonState = 0;          // variable for reading the pushbutton status

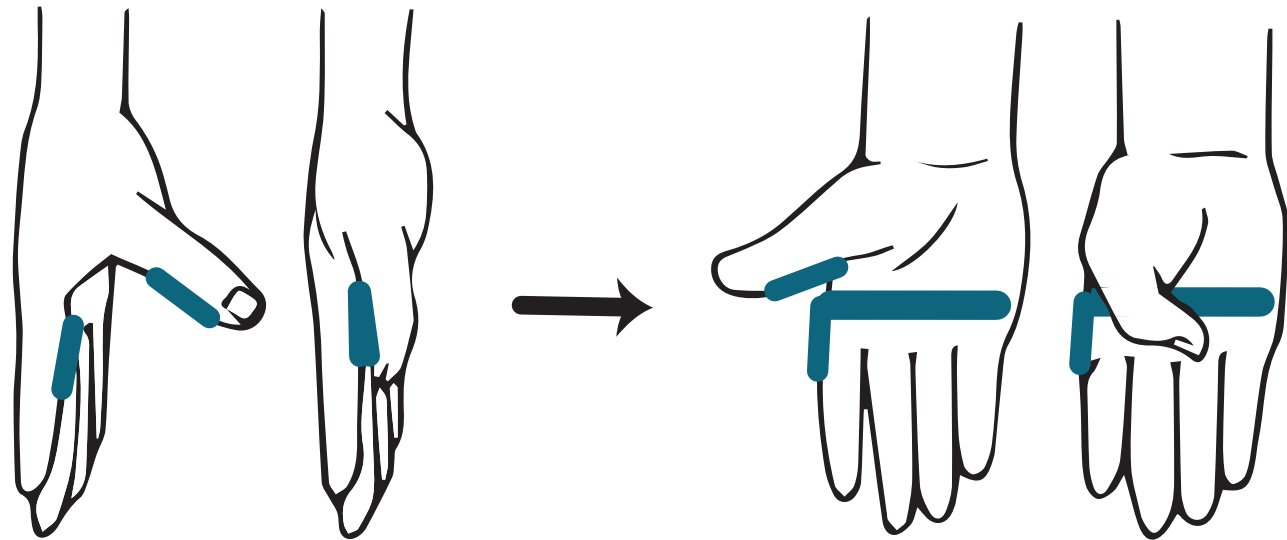
void setup() {
  pinMode(1, OUTPUT);
  pinMode(0, INPUT);
  pinMode(2, OUTPUT);
}

void loop() {
  // read the state of the pushbutton value:
  buttonState = digitalRead(0);

  // check if the pushbutton is pressed. If it is, the buttonState is HIGH:
  if (buttonState == HIGH) {
    for (int i = 0; i < 256; i++) {
      analogWrite(1, i); // PWM the LED from 0 to 255 (max)
      delay(1);
    }
    for (int i = 255; i >= 0; i--) {
      analogWrite(1, i); // PWM the LED from 255 (max) to 0
      delay(1);
    }
  }
  else {
    // turn LED off:
    digitalWrite(1, LOW);
  }
}
```

User Feedback

User driven development of the initial prototype.



The User

- University student, 20
- Confident cyclist
- Commutes by bike
- Cycles for leisure

Feedback 1

The user found their thumb did not consistently land on the conductive pad.

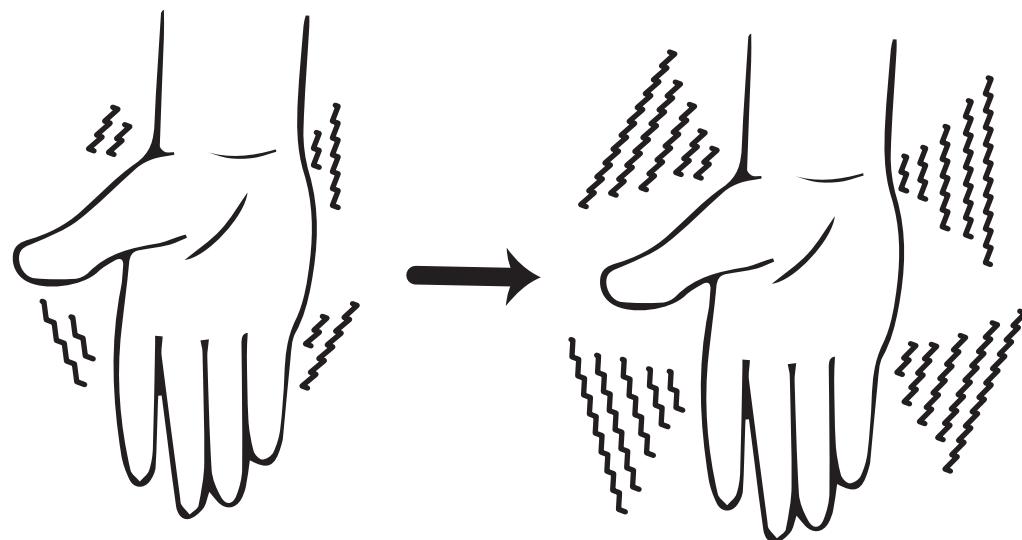
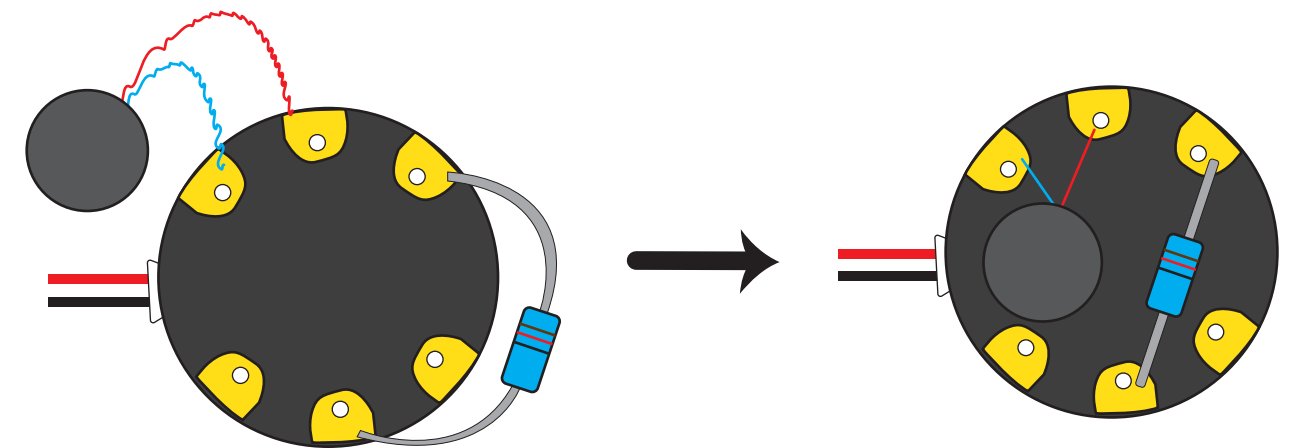
The user found the thumb actuation “easy to use” once they had practised several times. Their primary issue was that the conductive pad was too small and limited the number of hand positions that could be used to activate the lights.

An improvement would be to extend the amount of conductive area on the glove, possibly across the palm within the reach of the thumb.

Feedback 2

The user commented that the circuit “looked delicate” and was worried that they may damage it when putting on the glove.

The refined prototype will have the resistor soldered to the back of the board, and the vibration motor secured to the board instead of inside the glove. They suggested a piece of fabric inside the glove would make it more comfortable to wear.



Feedback 3

The user struggled to consistently identify the vibrations, as the pulses were “too short and quick”.

For the feedback session the glove was set to vibrate randomly within a 10 second period. Since there were only 2 vibrations for each ‘turn’ it was difficult for the user to catch the notification in time. The refined prototype will use a different pattern for the vibrations, with 4 pulses of longer duration. In the final product this could be customized within an app to the users preference.

Final Product & Brand DNA

Enjoy your journey with URSA guiding you

Wear URSA before starting journey



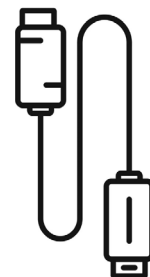
Turn device on and connect it to navigation system



If any of the available options feel too overwhelming, adjust them to your liking through the Settings tab on the app



Plug and recharge URSA through USB port



'Ursa' refers to the ursa minor, or little dipper, constellation. Constellations provide light to travelers, as do the LEDs attached to the product glove, and assist in their navigation, as does the integrated vibration motor.

The logo is a redesign of the shape of the constellation, with the north star being the biggest circle above the 'A'.



No screen. Be seen.

Our slogan: "No screen. Be seen." refers to:

Increasing cyclists' safety as they avoid looking down at their screens, away from the road, to navigate (the vibration motor tells them where to go)

Increasing cyclists' visibility by adding LEDs to the glove and enhancing their communication with surrounding drivers

Concept Validation

Testing our prototype in the real world.

[Link to Meeting Minutes and Gantt Chart](#)

Lights



- Lights are bright and well placed so are clearly visible at night.
- There are a variety of hand positions that the lights can be activated from.
 - Flashing lights increase visibility.



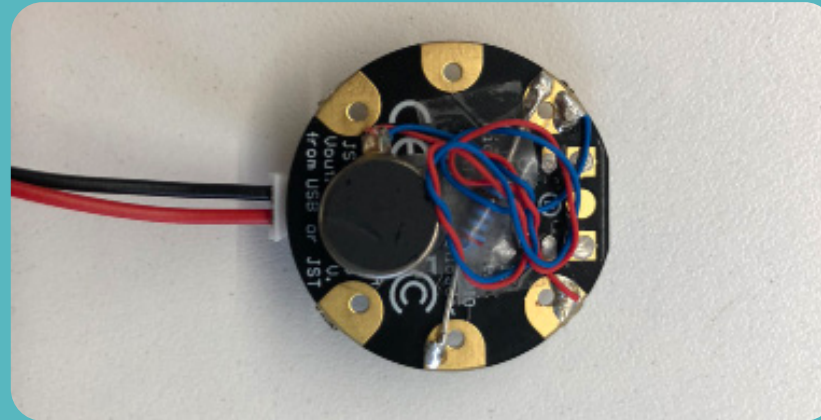
- Oncoming traffic cannot see the LEDs because they are on the back of the hand.

“

The thumb-index finger activation was easy to get the hang of.

”

Navigation



- Vibration is easy to interpret as it is separate for either hand and their respective direction.
- Vibration is strong enough to feel while cycling even on rougher surfaces.



- No interaction back to the smart device.
- App or extension required to customize.

“

Even on cycle paths which are more bumpy it is still easy to feel.

”

The Glove



- The conductive thread construction means minimal difference to a normal glove.
- Battery and circuit are concealed in the cuff of the glove and cannot be seen



- Improve material (waterproof, polyester)
- Move circuit or cover away from skin.

“

You can't feel the LEDs or wires but the board is scratchy on the skin.

”